

Corporate Finance PGE M1 Fall 2026

Jason Kang

Session 1: Introduction & Financial Statements

Jason Kang

- ▷ Assistant Professor in Finance
- ▷ PhD in Finance, INSEAD, 2026
- ▷ Office: B3.09
- ▷ Office Hours: E-mail for appointment or walk-in for short queries
- ▷ jason.kang@skema.edu

Arrive on time.

Don't cheat Edusign (automatic F). I will count # in classroom.

Electronic devices. No laptops. Tablets are allowed only for note taking.

General principle. Don't be a distraction.

About this class

Stop me if you need me to repeat.

Ask questions if you don't understand.

Participate in class discussions, and be respectful towards others.

- ▷ Overview of Corporate Finance
- ▷ Lecture Review
- ▷ IS, BS & CF Statements
- ▷ Time Value of Money?

SpaceX

+ Add to myFT

SpaceX sell-off wipes \$1tn from Elon Musk's rocket group

Stock trades below \$135 for first time since AI and space conglomerate's Wall Street debut in June

SpaceX stock dips below \$135 IPO price



Source: LSEG



Points raised:



“The subsequent slump in SpaceX’s stock price has come as shares in richly valued **tech stocks** have struggled over the past month over worries about **higher US interest rates** and the profitability of Big Tech’s **huge AI investments**.”

SpaceX sell-off wipes \$1tn from Elon Musk’s rocket group, Financial Times, 2026

Points raised:

- ▶ There is some relationship between interest rates and stock prices



“The subsequent slump in SpaceX’s stock price has come as shares in richly valued **tech stocks** have struggled over the past month over worries about **higher US interest rates** and the profitability of Big Tech’s **huge AI investments**.”

SpaceX sell-off wipes \$1tn from Elon Musk’s rocket group, Financial Times, 2026

Points raised:

- ▶ There is some relationship between interest rates and stock prices
- ▶ Tech stocks appear(?) to be affected more



“The subsequent slump in SpaceX’s stock price has come as shares in richly valued **tech stocks** have struggled over the past month over worries about **higher US interest rates** and the profitability of Big Tech’s **huge AI investments**.”

SpaceX sell-off wipes \$1tn from Elon Musk’s rocket group, Financial Times, 2026

Points raised:

- ▷ There is some relationship between interest rates and stock prices
- ▷ Tech stocks appear(?) to be affected more
- ▷ The profitability of investments (today) are also affected by interest rates

Example:

Suppose you have €10 million to invest in a company.

- ▷ **Option A:** SpaceX wants to use this money to invest in a new rocket launcher today. It expects to deliver profits (free cash flows) of €2mn starting from 10 years from now until forever.
- ▷ **Option B:** LVMH wants to use this money to build factories and expand production and train artisans to sell more handbags. It expects to deliver profits (free cash flows) of €5mn starting from 5 years from now until forever.

Which would you invest in?

Differences:

Corporate Finance is an introductory course to finance and its first principles.

In Corporate Finance, our goal is to make “good decisions”

We will focus mostly on risk and discount rates.

- ▷ Overview of Corporate Finance
- ▷ Lecture Review
- ▷ IS, BS & CF Statements
- ▷ Time Value of Money?

1. What is the main disadvantage of sole proprietorship compared to corporation?
2. What is a benefit corporation?
3. With a financial perspective, one can say that a firm is a pool of _____ in order to _____

4. What are the only two ways to externally finance a firm?

4. What are the only two ways to externally finance a firm?
- What influences the choice between debt vs equity?

5. How do we call the money brought by shareholders?
6. What is the difference between a shareholder and a debtholder?
7. What are the three financial statements and what are they used for? ?

- ▷ **Income Statement.** Captures financial performance: records revenues, expenses and profitability
- ▷ **Balance Sheet.** Captures financial health: fair value of assets (cash, inventory, PP&E), liabilities (debt)
- ▷ **Cash Flow Statement.** Keeps track of cash inflows/outflows to reconcile IS & BS

8. What is depreciation?

8. What is depreciation?

- ▷ Applies to physical assets
- ▷ Is a non-cash charge

8. What is depreciation?

- ▷ Applies to physical assets
- ▷ Is a non-cash charge
- ▷ Makes investing more attractive

Suppose you are a penniless CEO of a new start-up. To finance the purchase of new equipment for your business, you must raise €100.

- ▷ Compare the balance sheet & cash flow statement in Year 0 if you financed the purchase via equity versus debt.

Balance Sheet:

Year 0

Assets		Liabilities	
Cash	0	LT Debt	0
Fixed Assets	100		
		Shareholder's Equity	
		Paid-in Capital	100
Total Assets	100	Total L & SE	100

Balance Sheet:

Year 0

Assets		Liabilities	
Cash	0	LT Debt	100
Fixed Assets	100		
		Shareholder's Equity	
		Paid-in Capital	0
Total Assets	100	Total L & SE	100

Exercise 2.1

If a company incurs \$10 in depreciation expense, how does that affect the three financial statements?

Income Statement

Balance Sheet

**Cash Flow
Statement**

Exercise 2.2

What if a company makes a \$100 cash purchase of equipment? Assume straight line depreciation over 5 years, and a tax rate of 40%.

a. Financial Statements in Year 0

Income Statement

Balance Sheet

**Cash Flow
Statement**

Exercise 2.2

What if a company makes a \$100 cash purchase of equipment? Assume straight line depreciation over 5 years, and a tax rate of 40%.

a. Financial Statements in Year 1

Income Statement

Balance Sheet

**Cash Flow
Statement**

Exercise 2.3

Suppose the purchase (€100 at 5 yr straight-line depr / 40% tax) is financed via debt at a 10% interest rate.

a. Financial Statements in Year 1

Income Statement

Balance Sheet

**Cash Flow
Statement**

Exercise 1

- ▶ Mysmallbusiness company was incorporated in August. Money was raised at that time, equipment was purchased as well as frozen croissants .
- ▶ The company starts its operations on Sept 1, year n and goes until Dec 31. Over these 4 months, it sells for \$100 of croissants (warm not frozen) destocking from \$20 of its existing inventory.
- ▶ In order to inform Skema students about its new business, it makes and distributes flyers for a cost of \$8.
- ▶ On Dec 6, it decides to replenish its stock of frozen croissant and goes to Carrefour. Bus ticket costs \$2. It buys for \$30 of frozen croissants.
- ▶ You pay a 10% interest rate to your parents for the money they lent to you. Tax rate is 40
- ▶ All income and expenses are paid cash (no credit on sale)

Exercise 1.1

Assume no depreciation. Show the income statement for the period Sep 1 to Dec 31.

Income statement

Sales	100
--------------	------------

Cost of goods sold **20**

SG&A exp. 10

Depreciation

Operating income 70

Interest exp. 25

Earnings before tax	45
---------------------	----

Tax (40%) 18

Net income	27
------------	----

Exercise 1.1

Show the cash flow statement. (Net income = 27 and initial cash = 50.)

Show the cash flow statement. (Net income = 27 and initial cash = 50.)

Cash flow statement

from beg of Sep until end of Dec

Cash flow from operations

Net income 27

Additions to cash

Decrease in inventory	20
-----------------------	----

Subtractions from cash

Increase in inventory	30
-----------------------	----

Net cash from operations 17

Cash at the beginning of the period 50

Cash at the end of the period 67

Exercise 1.1

As of end of Aug			
Assets		Liabilities	
Fixed assets	450	Equity	300
		Common stock	300
Current assets	100		
Inventory	50	Long Term Debt	250
Cash	50		
Total assets	550	Total Liabilities	550

Exercise 1.1

As of end of Dec			
Assets		Liabilities	
Fixed assets	450	Equity	327
		Common stock	300
		Retained earnings	27
Current assets	127		
Inventory	60	Long Term Debt	250
Cash	67		
Total assets	577	Total Liabilities	577

Exercise 1.2

Assume a depreciation expense of \$10. Show the income statement for the period Sep 1 to Dec 31.

Exercise 1.2

Assume a depreciation expense of \$10. Show the income statement for the period Sep 1 to Dec 31.

Income statement	
As of end of Dec	
Sales	100
Cost of goods sold	20
SG&A exp.	10
Depreciation	10
Operating income	60
Interest exp.	25
Earnings before tax	35
Tax (40%)	14
Net income	21

Exercise 1.2

Cash flow statement & balance sheet?

Income Statement

Balance Sheet

**Cash Flow
Statement**

Exercise 1.3

- ▷ Same exercise but now we are said that the company sell \$80 cash and \$20 on credit
- ▷ It is offered a 4-month delay for paying the marketing expenses and taxes will be paid next fiscal year.
- ▷ On Dec 31, you decide to buy a second microwave oven financed with a \$150 one-year loan

Exercise 1.3

Income Statement

Balance Sheet

**Cash Flow
Statement**

Money in Different Currencies

Consider the following problem

$$€100 + \$100 = ?$$

Consider the following problem

$$€100 + \$100 = ?$$

Cannot add currencies without first converting into common currency

$$€100 + \$100 \times (0.86€/\$) = €186$$

$$€100 \times (\$1.16/€) + \$100 = \$216$$

Money at Different Points in Time

Would you rather receive €100 now or €100 in one year?

Would you rather receive €100 now or €100 in one year?

- ▷ **€100** now because I can consume it immediately.
Immediate consumption is preferred to deferred consumption.
= impatience (preferences)

Would you rather receive €100 now or €100 in one year?

- ▷ **€100** now because I can consume it immediately.
Immediate consumption is preferred to deferred consumption.
= impatience (preferences)
- ▷ **€100** now because I can grow it in one year.
I can deposit it at a bank at, for example, 5% interest rate, or invest in the stock market.
= opportunity cost

Would you rather receive €100 now or €100 in one year?

- ▷ **€100** now because I can consume it immediately.
Immediate consumption is preferred to deferred consumption.
= impatience (preferences)
- ▷ **€100** now because I can grow it in one year.
I can deposit it at a bank at, for example, 5% interest rate, or invest in the stock market.
= opportunity cost
- ▷ **€100** now because I might not receive **€100** next year for sure.
The future is risky.
= risk

A euro today is not the same as a euro in one year.

- ▷ The discount rate is an exchange rate that equalizes the value of money at different points in time.

A euro today is not the same as a euro in one year.

- ▷ The discount rate is an exchange rate that equalizes the value of money at different points in time.
- ▷ Finance is the valuation of cash flows that accrue in the future.

Today, we are going to use r to denote the discount rate / interest rate.

For now, we are going to ignore risk entirely.

Suppose the interest rate is 5%. What is the value of €100 in one year if you deposit it in a bank?

Suppose the interest rate is 5%. What is the value of €100 in one year if you deposit it in a bank?

For a single cash flow:

$$\text{Future Value (FV)} = C \times (1 + r)^t$$

where

- ▷ C = cash flow
- ▷ r = discount rate
- ▷ t = time

Suppose the interest rate is 5%. You need €105 in one year.
How much money do you deposit in a bank **today** to make sure
you have €105 one year from now?

Suppose the interest rate is 5%. You need €105 in one year.
How much money do you deposit in a bank **today** to make sure you have €105 one year from now?

For a single cash flow:

$$\text{Present Value (PV)} = \frac{C}{(1 + r)^t}$$

Searching for r

What is the discount rate that makes you indifferent between receiving €100 today versus €105 in one year?

Example

Suppose $r = 7\%$. You need to buy a copier that costs \$9,500.
You can either buy it today, or buy it next year at \$10,000.
Which do you prefer?

Example

The cost of renovating a bridge was \$3bn in 2004. At the time, engineers estimated that if the project were delayed to 2005, the cost would rise by 10%. Assume $r = 2\%$. What is the cost of delay in terms of dollars in 2004?

Hint:

- Find the cost of doing the project in 2005
- Bring this cost to present value
- Compare this to the cost of doing the project in 2004.

Example

The cost of renovating a bridge was \$3bn in 2004. At the time, engineers estimated that if the project were delayed to 2005, the cost would rise by 10%. Assume $r = 2\%$. What is the cost of delay in terms of dollars in 2004?

Example

Your firm has a risk-free investment opportunity where it can invest \$160,000 today and receive \$170,000 in one year. For what level of interest rates is this project attractive?